3W Well Data Viewer

A desktop GUI to inspect the real instances of the 3W dataset

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Why a viewer for the real instances?

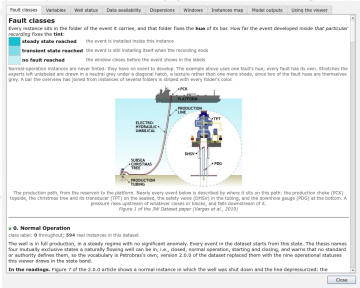
- 3W 2.0.0 ships **1,119 real instances** on **40 wells**; 986 of them overlap another instance of their well in time. One stretch of a well can be filed twice, under two labels.
- Every file carries every column the dataset declares. **What each sensor actually recorded** is a question of its own: the downhole pressure is frozen in more than half of the real instances.
- The literature answers such questions with figures **drawn by hand**, one well or one fault at a time, from a sample of the instances.
- The viewer draws them **live, over every real instance**, the time series of any instance one click away. It trains no model: it is a lens on the data as 3W ships it.

Drawn by hand in the literature	Drawn live, on the page
The instances of a well along time	**Timelines**
Availability maps of the sensors	**Availability**
Instances of one fault side by side	**Faults, Features**
One variable against another	**Dispersions**
Fixed-size, one-label windows, as a model reads them	**Windows**
Clusterings and one-class models of the instances	**Instances**
The Toolkit's `CleanSignals` rule	**Availability, Timelines**

Seven pages, one question each

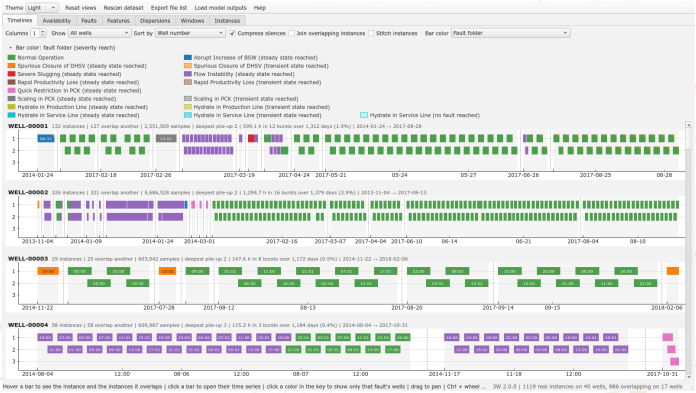
Page	Question
Timelines	How do the instances and the fault classes of each well spread out across time?
Availability	What did each sensor actually record, beyond what the file declares?
Faults	How does one fault look on every well it occurred on?
Features	What does one sensor read under every fault class?
Dispersions	How do two sensors move together, and how much of it is the historian's interpolation?
Windows	What does a model see: one sensor of a well in fixed-size windows, one label each?
Instances	Where does every real instance sit on one plane, by what its sensors amount to?

A click on any bar, cell, point or plot opens the **instance window** of that instance and of every instance it overlaps. `F1` anywhere opens the Help.



The Help: the platform and its sensors, the fault classes and their signatures, the well statuses, the data availability, the sources.

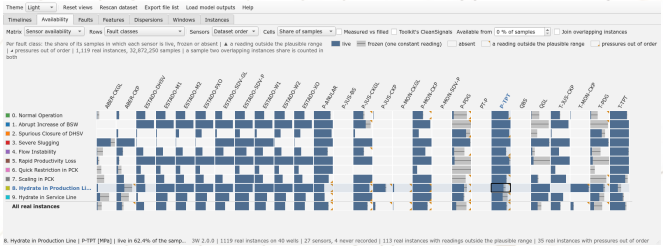
Timelines: every real instance of every well

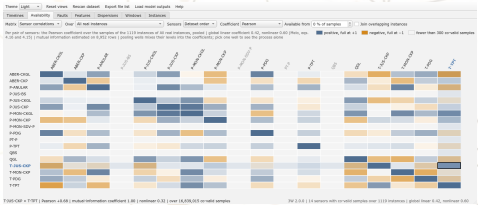


- One timeline per well; every instance a bar from its first to its last timestamp, **stacked where it overlaps** another.
- **Join overlapping instances** merges the bars whose labels agree, so what still overlaps is exactly the labeling conflicts. **Stitch** lays a well's recordings end to end.
- **Bar color**: the fault folder, a sensor's availability or measurements, a descriptor, what the Toolkit's `CleanSignals` keeps, the agreement with model outputs.
- Hover names an instance and its partners; a click opens the instance window.

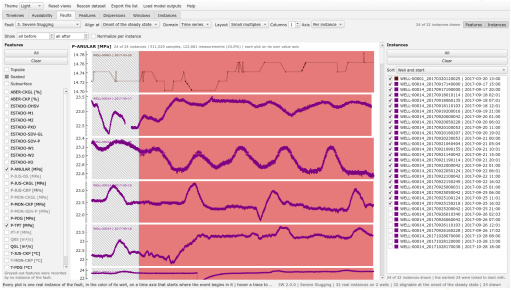
Availability: what each sensor actually recorded

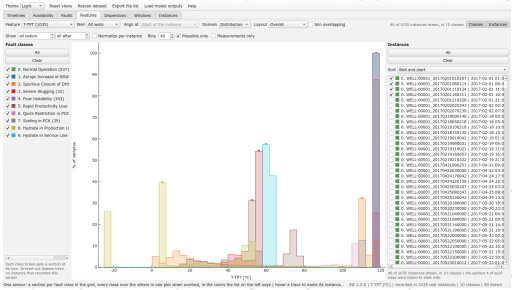
- **Sensor availability**: per sensor and group of instances, the share **live**, **frozen** or **absent**. A frozen gauge is not available.
- **Sensor pairs**: how often two sensors carry a reading at the same instant. 102 of the 351 pairs never do.
- **Sensor correlations**: Pearson, mutual information and the nonlinear part, summed into Melo's global coefficient.
- The Toolkit's `CleanSignals` applied on the fly, thresholds movable: the cells it would discard are slashed, the columns it would drop greyed.
- **Corner marks**: an implausible reading (amber), pressures out of order (amber), a well status its valves contradict (red).





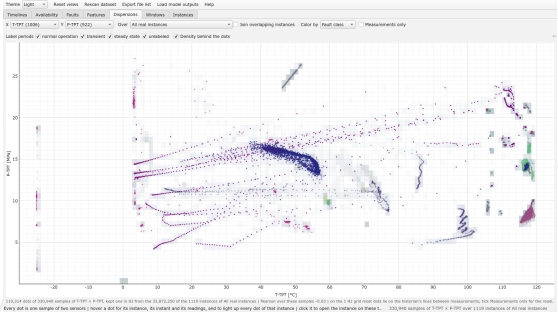
Faults and Features: across wells, across events

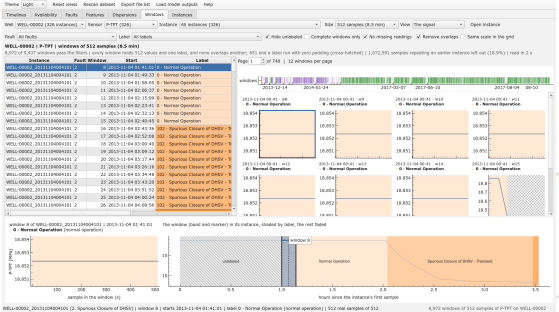




- **Faults**: every real instance of one fault, from every well, **aligned at the onset** of the event, one section per sensor, one color per well.
- **Features**: the question turned round: one sensor, one section per fault class, so what a gauge reads under a hydrate sits beside what it reads under slugging.
- Both: small multiples or overlaid; the **time**, **distribution** or **spectrum** domain (Welch on the grid, Lomb-Scargle on the measurements alone); z-scores per instance; the label periods shaded behind every trace.

Dispersions and Windows: two more views

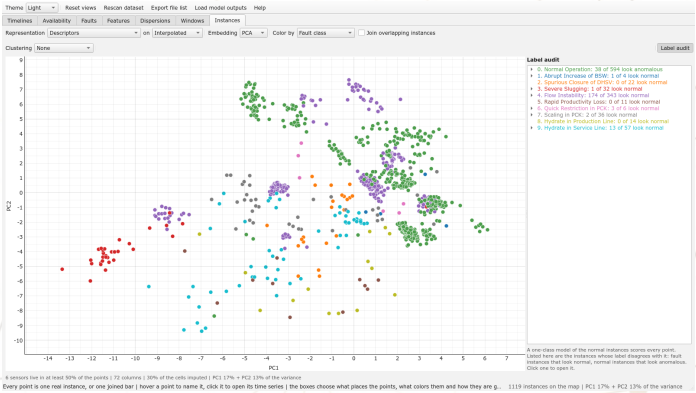




- **Dispersions**: two sensors against each other, every sample a dot over its density, colored by class, well or label period.
- **Measurements only** keeps the instants at which both were read: under severe slugging P-TPT against T-TPT reads $+0.40$ over every sample and $+0.95$ over the 5% measured.

- **Windows**: one sensor of a well cut as a model reads it: 256, 512 or 1024 samples a window, **one label each**, zero padded, no instant twice.
- A statistic per window and how well it **separates** normal from event windows; the largest well, 5.7 million samples, is cut in two seconds.

Instances: every real instance on one plane



- 1,119 points. An instance becomes a point by the **descriptors** of its sensors (moments, quantiles, autocorrelation time, signal-to-noise ratio, Gaussianity), by their shape alone, or by the Toolkit's **DTW** distance.
- **PCA**, **t-SNE** or **UMAP**; k-means, Gaussian mixture, agglomerative or DBSCAN, scored against the fault classes and against the wells: is it the event or the well that places an instance?
- **Typicality**, the distance to the medoid of the class, and a **label audit**: a one-class model of the normal instances lists the fault instances that look normal and the normal ones that look anomalous.

The instance window: traces, labels, warnings



- One block per instance, or per joined recording, on a **shared time axis**: the well status and the label as bands, a plot per sensor, the **measurements as dots** on the historian's line.
- Three warnings, from a survey of every real instance of 3W 2.0.0:
  - **amber**: a reading no instrument could produce (a negative pressure, 30,000 °C);
  - **amber, dashed**: two pressures of one line out of order: **35 of 1,119** instances;
  - **red**: a well status its valves contradict, Open with the wing valve closed: **41 of 1,119** instances.

What the viewer found: the historian's lines

6 %
of the samples of 3W 2.0.0 are measurements

55 % + 39 %
interpolated and held between them by the PI historian

33 s
median interval between two measurements

- The dataset is sampled once a second, but the sensors were not read once a second: the historian **interpolated** between the values it archived. Melo saw it on WELL-00001 (doctoral thesis, section 4.2.5); the viewer tests every sample of every real instance.
- The rule: a sample equal to the one before is **held**; one collinear with both its neighbours is **interpolated**; everything else is a **measurement**. The tolerance, one part in a million, is read off the data.
- Every figure taken on the 1 Hz grid inherits the lines. Wherever the viewer shows one it says so, and offers it on the **measurements alone**.

T-JUS-CKP	on the grid	measured
signal-to-noise ratio	36,000	1.5
autocorrelation time	7 min	73 s

Where live, P-MON-CKP is read every 12 s, P-PDG every 13 s, T-TPT every 16 s, QGL every 21 s, P-TPT every 100 s, T-JUS-CKP every two minutes and P-ANULAR every four.

Meeting the 3W Toolkit

PETROBRAS

By rule

The Toolkit's `CleanSignals` decides, per instance and sensor, whether a signal is to be believed. The viewer holds exactly what the rule needs, so it applies it at no cost, lets its thresholds move, and marks every cell, bar and block the rule would discard, saying why.

By file

Export file list writes the instances a page has on show as the JSON of a `ParquetDatasetConfig` with `split="list"`, which the Toolkit loads with one line. The provenance is written beside it.

By model outputs

A folder: `model.json` and one parquet file per instance scored, with a `timestamp` index and a `label` column. The **agreement** with the experts' labels is drawn as a third band in the instance window, a bar color and a map coloring.

- The example shipped: a PCA control chart over the 319 real instances of four wells, one model per well. It agrees with the labels 98 % of the time on the normal instances and **between 100 % and 5 %** on the fault instances, the same event reading 100 % on one well and 5 % on another: the agreement says as much about a well's normal data as about the event.
- What cannot come back: the Toolkit's `predictions_<timestamp>.csv` carries no instance and no instant, so nothing in it can be drawn onto the data. The folder format is the smallest one that says both.

What the viewer says back to 3W

41

real instances whose well status the valves contradict

35

real instances with two pressures of one line out of order

1.6 %

unlabeled in the largest joined recording, a third apart

- **Labels worth a second look.** 41 instances are labeled Open while a valve of the production path reads closed or a crossover open, mostly for the whole recording; 35 carry a downstream pressure above its upstream one. Each is marked on the pages and one click from its trace.
- **Overlapping windows are one recording.** Joined, the largest group of 3W 2.0.0 (71 windows over six days) keeps 1.6 % of its samples unlabeled where the windows apart left a third: the merged recording is what a model would train on.
- **The 1 Hz grid is the historian's, not the sensors'.** A spectrum, an autocorrelation or a signal-to-noise ratio taken on the grid describes the interpolation as much as the process; the measurements, 6 % of the samples, say which is which.
- **A model export that names the instance and the instant** can be drawn onto the data and compared with the experts, instance by instance. The viewer's folder format is one proposal for it.

Try it

- Python 3.11 or newer, uv and a local copy of 3W 2.0.0; at least 1.4 GB of RAM. MIT license.
- The first launch scans the 1,119 instances in about 7 s and caches the catalogue; the heavier passes run once, behind a progress dialog, and are cached too.
- `backend` (pandas, numpy, pyarrow), `algorithms` (numpy) and `frontend` (PySide6, pyqtgraph); the heavier analyses are optional extras.

From a shell

```
git clone https://github.com/Thiago-TP/3w-well-data-viewer
cd 3w-well-data-viewer
uv sync --all-extras
uv run overlap-viewer --raw-dir /path/to/3W/dataset
```

github.com/Thiago-TP/3w-well-data-viewer

Issues, questions and pull requests are welcome.



Thank you!



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